ADL Aim and Theory

Experiment 1 – Implement multilayer perceptron algorithm for MNIST Hand-Written Digit Classification.

A Multilayer Perceptron (MLP) is a type of artificial neural network made up of interconnected neurons arranged in input, hidden, and output layers. It is widely used for classification problems. In this experiment, the MNIST dataset containing handwritten digit images is used to train the network to recognize digits from 0 to 9. The model learns by adjusting weights through backpropagation and gradient descent. Activation functions such as ReLU and Softmax improve learning efficiency and multiclass classification performance.

Experiment 2 – Design a neural network for classifying movie reviews (Binary Classification) using IMDB dataset.

This experiment focuses on sentiment analysis using the IMDB movie review dataset. A neural network is trained to classify reviews as positive or negative based on the textual content. Since neural networks cannot directly process text, preprocessing techniques such as tokenization, vectorization, and padding are applied. The model identifies patterns in frequently occurring words and learns sentiment-related features. Sigmoid activation and binary cross-entropy loss are commonly used for binary classification tasks in deep learning.

Experiment 3 – Design a Neural Network for classifying news wires (multi-class classification) using Reuters dataset.

The Reuters dataset consists of news articles divided into multiple categories such as politics, business, and sports. In this experiment, a neural network is designed for multiclass text classification. The text data is first converted into numerical vectors using encoding methods. Hidden layers help the model learn meaningful relationships between words and categories. The output layer uses Softmax activation to predict the probability of each class, while categorical cross-entropy is used as the loss function.

Experiment 4 – Design a neural network for predicting house prices using Boston Housing Price dataset.

This experiment demonstrates the use of neural networks for regression tasks. The Boston Housing dataset contains features such as crime rate, average number of rooms, and property tax values to predict housing prices. Unlike classification, regression predicts continuous numerical values. The neural network learns relationships between input features and house prices during training. Mean Squared Error (MSE) is commonly used as the loss function, and performance is evaluated based on prediction accuracy.

Experiment 5 – Build a Convolution Neural Network for MNIST Handwritten Digit Classification.

A Convolutional Neural Network (CNN) is a deep learning architecture specially designed for image processing and computer vision tasks. CNN automatically extracts important features from images using convolution and pooling operations. In this experiment, the MNIST handwritten digit dataset is used for classification. Convolution layers detect patterns such as edges and shapes, while pooling layers reduce dimensionality. Fully connected layers perform final classification. CNN provides higher accuracy and better feature extraction compared to traditional neural networks.

Experiment 6 – Build a Convolution Neural Network for simple image (dogs and Cats) Classification

This experiment uses a CNN model to classify images of dogs and cats. The network learns visual patterns such as fur texture, facial structure, and object shapes from training images. Convolution layers extract image features, while pooling layers reduce computational complexity. Data augmentation techniques like rotation and flipping may be used to improve generalization and reduce overfitting. The final output layer predicts whether the given image belongs to the dog or cat category.

Experiment 7 – Use a pre-trained convolution neural network (VGG16) for image classification.

VGG16 is a popular pre-trained convolutional neural network developed for large-scale image classification tasks. It is trained on the ImageNet dataset containing millions of images. In this experiment, transfer learning is applied by reusing the learned features of VGG16 for a new classification problem. This approach reduces training time and improves accuracy, especially when limited training data is available. Additional layers can be added and fine-tuned according to the target application requirements.

Experiment 8 – Implement one hot encoding of words or characters.

One Hot Encoding is a preprocessing technique used to convert categorical textual data into numerical form for machine learning and deep learning models. In this method, each word or character is represented as a binary vector where only one element is set to 1 and all others are set to 0. This representation helps neural networks understand textual information mathematically. One hot encoding is widely used in Natural Language Processing tasks such as text classification and language modeling.